Blockchain-Based Crowdfunding Platform

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STATEMENT OF SUBMISSION

This is to certify that following students have successfully completed the final project named as: _____ at University of the Punjab, Gujranwala Campus, Gujranwala, to fulfill the partial requirement of the degree of **Bachelor in Computer Science.**

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Write to us

I welcome your response to this project. If there is anything you want to mention about the improvement of this project please let us know:

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ABSTRACT

This project introduces a decentralized crowdfunding platform developed on the Ethereum blockchain, aiming to deliver a transparent, secure, and efficient alternative to conventional crowdfunding systems. By utilizing blockchain technology, the platform enables users to create campaigns with specific goals and timelines, donate securely using Ethereum wallets, track progress in real-time, and withdraw funds automatically upon goal completion. All transactions are immutably recorded on the blockchain, ensuring full transparency and trust for both creators and donors.

The technical implementation features a responsive frontend built with React, paired with a backend powered by Solidity smart contracts deployed on the Holesky testnet. Integration with the blockchain is facilitated by the Thirdweb SDK, while campaign content is stored decentrally using IPFS, enhancing resilience and accessibility. The platform's feasibility is supported by mature blockchain tools, addressing the rising demand for decentralized finance (DeFi) solutions. Economically, it offers a low development cost of $15,000 and a projected return on investment of 844% within the first year, underscoring its viability and scalability.

This Ethereum-based crowdfunding solution redefines fundraising by providing a decentralized, trustworthy ecosystem, empowering users globally to support innovative projects with confidence and efficiency.

For My Beloved Parents and Teachers

"Dedicated to my parents and teachers without whose wholehearted support, encouragement and guidance it would have been impossible for me to make this project"

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CHAPTER # 1

1.1 Project/Product Feasibility Report

1.1.1 Technical Feasibility

The project is technically feasible due to the availability of mature blockchain frameworks like Ethereum and tools such as Thirdweb SDK, which streamline smart contract development and integration. The team has expertise in React, Solidity, and blockchain technologies.

Challenges: Ensuring smart contract security and managing blockchain-specific issues such as gas fees and transaction delays.

1.1.2 Operational Feasibility

The platform **is** partially operational due to current level of awareness and lack of current legal approvals in Pakistan. However, the platform aligns with the rising demand for decentralized finance (DeFi) solutions, making it operationally viable. It will cater to tech-savvy users and blockchain enthusiasts with an intuitive interface and proper onboarding.

Challenges: Educating non-blockchain users on wallet setup and cryptocurrency transactions.

1.1.3 Economic Feasibility

Development costs are minimized by using the Holesky testnet and open-source tools like React and the Thirdweb SDK, keeping the initial investment low at an estimated $15,000, which includes developer salaries, testing, and deployment. Long-term revenue can be generated through a 2% transaction fee on successful campaigns and premium features like promoted listings, projecting a monthly revenue of $12,000 (based on 100 campaigns per month with an average funding of $5,000 each, yielding $10,000 from fees, plus $2,000 from premium features). Operational costs, such as IPFS hosting and blockchain transaction fees, are minimal at $200 per month, resulting in a net monthly profit of $11,800.

- Cost-Benefit Analysis: The $15,000 development cost is dwarfed by the projected monthly profit of $11,800, offering a strong benefit-to-cost ratio. The use of cost-effective tools and scalable revenue streams ensures significant financial upside.
- Break-Even Point: The project will recover its initial $15,000 investment within two months of launch, as cumulative revenue exceeds costs by the end of month two.
- Return on Investment (ROI): With an annual profit of $141,600 ($11,800 × 12), the ROI is approximately 844% in the first year, showcasing exceptional economic potential.

This analysis confirms a positive outlook, with rapid cost recovery and robust long-term profitability.

Challenges: Cryptocurrency price volatility could affect donation amounts and user adoption.

1.1.4 Schedule Feasibility

The project can be completed within the timeline (January 25, 2025 - May 1, 2025) using an agile approach and reusable components.

Challenges: Delays in smart contract auditing or integration testing could impact the schedule.

1.1.5 Specification Feasibility

Core requirements (campaign creation, donation, withdrawal) are clear and manageable within the timeline. Smart Contract is manageable and deployable using Thirdweb SDK.

Challenges: Avoiding scope creep from additional features like multi-currency support.

1.1.6 Information Feasibility

Recent delaying strategies for donating funds to cities like Gaza and Kashmir proves that it is the need of the hour to have such a donation system which is uncensored as well as anonymous (to save donators from hatred and attacks). Moreover, sufficient information is available from blockchain documentation, Thirdweb SDK resources, and user feedback during beta testing.

Challenges: Keeping up with evolving blockchain regulations.

1.1.7 Motivational Feasibility

Having an intense interest in AI and Blockchain, this project is aimed to be one of the most memorable and interesting since it solves the centuries-long stopping hands from donating problem. So, the team is highly motivated to make a difference.

Challenges: Sustaining motivation during complex blockchain integration phases especially after seeing the current treatment by the nation which needs it the most.

1.1.8 Legal & Ethical Feasibility

The blockchain-based projects have recently received approvals by Government of Pakistan. Moreover, cryptocurrency being legal makes this project fully legal in Pakistan. However, a few countries still don't accept cryptocurrency but there is a high chance that they will accept it soon. Ethically, this project earns a lot of respect from the ones who want to help people even when their own country stands against it. The project adheres to legal standards by using the Holesky testnet for development. Ethical considerations include user privacy and secure fund management.

Challenges: Addressing cryptocurrency regulations across different regions.

1.2 Project/Product Scope

The Product scope includes development of a decentralized crowdfunding platform using Ethereum smart contracts. There must be user authentication via Ethereum wallets. The core functionalities of this product include campaign creation, donation, monitoring, and withdrawal. The real-time campaign progress tracking is also included. It is important to host this product only on a decentralized hosting platform such as Fleek which provides decentralized hosting.

The out-of-scope features include Fiat currency support, tracking real-world identity of the donors, and anything that can compromise the integrity of being anonymity or decentralization. Multi-blockchain integration beyond Ethereum is not included in product scope due to their less stability.

CHAPTER # 2

1.3 Project/Product Costing

Unadjusted Function Points (UFP) Table

Components	DET	RET/FTR	Complexity	FP
ILF - Campaign Data	6	1	Low	7
ILF - User Data	2	1	Low	7
EIF - Ethereum Blockchain	1	1	Low	5
EIF - Thirdweb SDK	1	1	Low	5
EI - Campaign Creation	6	1	Average	4
EI - Donation	2	2	Average	4
EI - Withdrawal	3	2	Average	4
EO - Campaign List	5	1	Low	4
EO - Donation Receipts	3	1	Low	4
EO - Withdrawal History	4	1	Low	4
EQ - Campaign Search	2	1	Low	3
EQ - User Profile	3	2	Average	4

Total UFP: 59

Explanation of Values

1. **DET**: Number of unique data fields (e.g., Campaign Data has title, description, target, deadline, image, owner = 6 DET).
2. **RET/FTR**: Number of record types or files referenced (e.g., Donation EI updates Campaign ILF and reads User ILF = 2 FTR).
3. **Complexity**: Determined using FPA tables (e.g., EI with 2 DET and 2 FTR is Average).

Complexity Adjustment Factors (Fi) Table

SR.#	QUESTION	SCALE
1	Does the system require reliable backup and recovery?	2
2	Is data communication required?	4
3	Are there distributed processing functions?	3
4	Is performance critical?	3
5	Will the system run in an existing, heavily utilized operational environment?	1
6	Does the system require online data entry?	4
7	Does the online data entry require the input transaction to be built over multiple screens or operations?	1
8	Are the master files updated online?	4
9	Are the inputs, outputs, files, or inquiries complex?	3
10	Is the internal processing complex?	2
11	Is the code designed to be reusable?	4
12	Are conversion and installation included in the design?	1
13	Is the system designed for multiple installations in different organizations?	2
14	Is the application designed to facilitate change and ease of use by the user?	4

ΣFi: 38

Explanation of Ratings

1. **Backup and Recovery (2)**: Blockchain is immutable, but local data may need backup.
2. **Data Communication (4)**: Required for blockchain and IPFS interactions.
3. **Distributed Processing (3)**: Blockchain is distributed, but the app itself is not.
4. **Performance (3)**: App should be responsive despite blockchain delays.
5. **Operational Environment (1)**: Runs in a browser, not heavily constrained.
6. **Online Data Entry (4)**: Essential for forms.
7. **Multiple Screens (1)**: Forms are simple, single-page.
8. **Master Files Online (4)**: Updated via blockchain.
9. **Complexity (3)**: Moderate complexity in inputs/outputs.
10. **Internal Processing (2)**: Smart contract logic is external.
11. **Reusability (4)**: React components are reusable.
12. **Conversion/Installation (1)**: Minimal for a web app.
13. **Multiple Installations (2)**: General use, not organization-specific.
14. **Ease of Use (4)**: User-friendly design is a priority.

Calculation of Function Point (FP):

$FP_{estimated}$ = Count Total * (0.65 + 0.01 * Σ(Fi)
= 59 (0.65 + 0.01 * 38)

= 60.77 (Total Adjusted FP)

Total Duration of The Project:

Average productivity = 12 FP / Person-month

Effort month = $FP_{estimated}$ / Productivity
= 60.77 / 12

= 5.06 / Person-month

Duration of Project = Effort month / No. of Persons
= 5.06 / 1

= 5.06 months (approx.)

Total Cost of The Project:

Labor Rate = 120,00 PKR

Cost Per FP = Labor Rate / Avg. Productivity
= 120,000 / 12

= 12,000 PKR

Total Project Cost = $FP_{estimated}$ * Cost Per FP

Total Project Cost (Estimated) = 730,000 PKR (approx.)

CHAPTER # 3

3.1 CPM - Critical Path Method

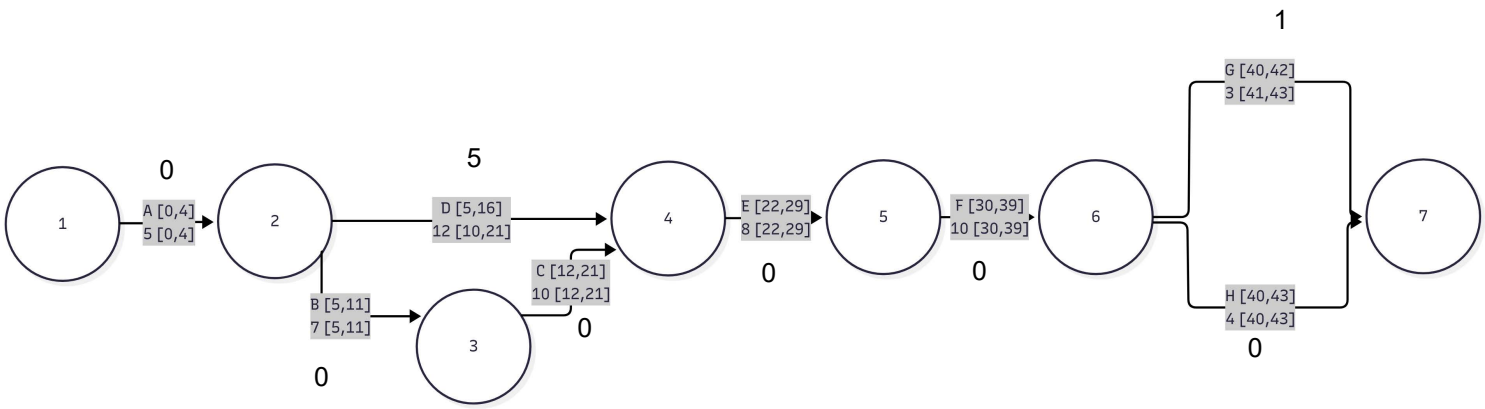
Task Estimated Completion Table

Task	Task Description	Duration (days)
A	Requirement Gathering and Analysis	5
B	UI/UX Design	7
C	Frontend Development (React Components)	10
D	Smart Contract Development (Solidity)	12
E	Blockchain Integration (Thirdweb SDK)	8
F	Testing (Unit, Integration, Security)	10
G	Deployment (IPFS, Domain Setup)	3
H	User Documentation and Training	4

Task Dependency Table

Task	Immediate Predecessor	Duration (days)
A	None	5
B	A	7
C	B	10
D	A	12
E	C, D	8
F	E	10
G	F	3
H	F	4

3.1.1 Network Diagram of CPM



Explanation:

1. Task A starts the project.
2. Task B follows A.
3. Task C follows B.
4. Task D follows A.
5. Task E requires both C and D.
6. Task F follows E.
7. Tasks G and H both follow F, leading to the project's end.

3.1.2 Estimate activity completion Time

This table includes calculations for Early Start (ES), Early Finish (EF), Late Start (LS), Late Finish (LF), Total Slack (TS), and Free Slack (FS) for each task, following the CPM methodology.

Task	Duration	ES	EF	LS	LF	TS	FS
A	5	0	4	0	4	0	0
B	7	5	11	5	11	0	1
C	10	12	21	12	21	0	1
D	12	5	16	10	21	5	6
E	8	22	29	22	29	0	1
F	10	30	39	30	39	0	1
G	3	40	42	41	43	1	0
H	4	40	43	40	43	0	0

Note:

- **ES and EF**: Calculated via forward pass (ES + Duration - 1 = EF).
- **LS and LF**: Calculated via backward pass, with LF of end tasks (G and H) set to the project completion time (43).
- **TS**: LF - EF, indicating delay tolerance.
- **FS**: Minimum ES of successors - EF, showing flexibility before affecting successors.

3.1.3 Identify Critical Path(s)

The critical path is the sequence of tasks with zero total slack (TS = 0), as any delay in these tasks delays the entire project.

Critical Path: **A → B → C → E → F → H**

Project Duration: 43 days
Path Breakdown:

- A: 0 to 4 (5 days)
- B: 5 to 11 (7 days)
- C: 12 to 21 (10 days)
- E: 22 to 29 (8 days)

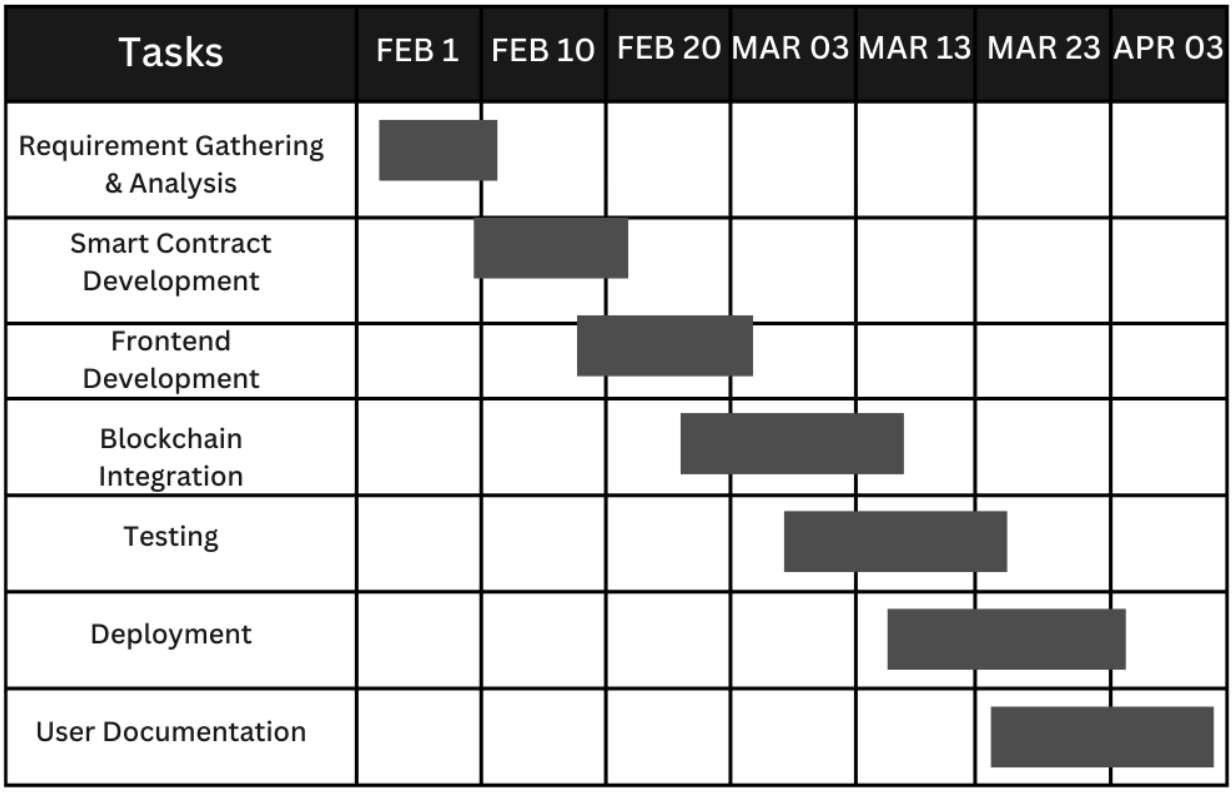
- F: 30 to 39 (10 days)
- H: 40 to 43 (4 days)

Total duration aligns with the project completion on day 43.

Other Paths:
- A → D → E → F → H
- A → B → C → E → F → G)

Both include tasks with slack (D: 5 days, G: 1 day), so they are not critical.

3.2 Gantt chart



3.3 Introduction to Team member and their skill set

- **Project Leader: [Muhammad Awais] -** Skilled in blockchain development and project management.
- **Blockchain Developer: [Muhammad Awais] -** Proficient in Solidity and smart contract development.
- **Frontend Developer: [Muhammad Awais] -** Experienced in prompt engineering to create UI/UX.
- **QA Engineer: [Muhammad Awais] -** Expert in testing decentralized applications.

3.4 Tools and Technology with reasoning

Front-End Development

- **React**: Component-based framework for a scalable, reusable UI.
- **Thirdweb SDK**: Simplifies blockchain integration for efficient development.

Back-End Development

- **Solidity**: Enables secure, reliable smart contracts on Ethereum.
- **Ethereum (Holesky)**: Testnet for cost-effective development and testing.

Development and Testing Tools

- **Remix:** Contract call and application integration during testing for smart contracts without deployment.

Project Management

- **Git and GitHub**: Version control and team collaboration.

Host Platform

- **Fleek:** Hosting platform allowing decentralized hosting and CID usage instead of standard IP-based hosting.

3.5 Vision Document

The decentralized crowdfunding platform is an innovative blockchain-based solution designed to revolutionize fundraising by offering a transparent, secure, and efficient alternative to traditional crowdfunding systems. Built on the Holesky testnet and integrated with the Thirdweb SDK, this platform enables users to create, manage, and donate to campaigns using cryptocurrency (ETH), ensuring decentralized access and trust through immutable ledger technology. The frontend, developed with React, provides a responsive and user-friendly interface, while the platform's hosting on IPFS via Fleek—accessible at dominate.pw—guarantees resilience and decentralization. Key functionalities include campaign creation with input validation (e.g., image URL verification), real-time tracking of funding progress, and secure mechanisms for fund withdrawals, all underpinned by smart contracts. The platform aims to minimize development costs by leveraging open-source tools and the Holesky testnet, with projected revenue streams from transaction fees and premium features. Economically, it promises a remarkable 844% ROI in its first year, reaching a break-even point within two months, making it a financially viable endeavor. Development is scheduled to begin on February 1, 2025, with a critical path of 43 days encompassing requirement gathering, UI/UX design, frontend and smart contract development, blockchain integration, testing, deployment, and user documentation. By harnessing blockchain's transparency and security, this platform not only meets modern fundraising needs but also empowers users with a decentralized, trustworthy system for supporting causes and projects worldwide.

3.6 Risk List

Risk	Impact	Probability	Mitigation Strategy
Smart contract vulnerabilities	High	Medium	Conduct thorough audits and testing.
User adoption challenges	Medium	High	Provide educational resources and support.
Cryptocurrency volatility	Medium	High	Consider stablecoin support in future updates.
Regulatory compliance	High	Low	Monitor legal developments and adapt.
Development delays	Medium	Medium	Use agile methods and include buffer time.

CHAPTER # 4

4.1 Requirement Engineering

4.1.1 System Specifications

The system specifications of both client and server are as follows.

4.1.1.1 Hardware Requirements

Server-Side

Hardware	Requirement	Reason
RAM	8 GB	Core system applications of Windows 11 use at least 4GB RAM plus extra free RAM is required for working on React server.
Storage	120 GB Harddrive Or 5 GB (Free-space)	React modules and Thirdweb SDK take at least 3 GB storage plus at least 2 GB extra space is required for cache.

Client-Side

Hardware	Requirement	Reason
RAM	2 GB Or 100 MB (free)	User's entered data is temporarily stored on RAM before passing to server.
Storage	1 MB (free-space)	User's entered data is temporarily stored in cache while making requests to the server.

4.1.1.2 Software Requirements

Server-Side

Software	Choice	Alternatives	Reason
Operating System	Windows 11	Linux, MacOS	Thirdweb SDK is currently stable only on the latest Windows systems.
Thirdweb SDK	V4 or V5	Infura	Handles blockchain interaction with the application more efficiently and supports Holesky, stable testnet blockchain which Infura doesn't.
Smart Contract	Solidity	Vyper	Solidity provides more authentication checkpoints during contract call while focuses on reducing gas fee and thus less authentications.
React	19	Angular	React provides optimized performance for applications using Thirdweb while Angular slows down blockchain interactions.

Client-Side

Software	Choice	Alternatives	Reason
Operating System	Windows	Linux, MacOS	Windows offers support for wide variety of COLD wallets. Moreover, Client-side requires only browser and wallet to interact with the application.
Browser	Brave Browser	Chrome, Edge	Brave Browser has support for Web3-based applications like built-in stable wallet, support for Web3-based TLDs.
Web3 Wallet	Brave Wallet	MetaMask, Trust	Brave Wallet provides dual signature authentication option to verify transactions while Metamask is frequently compromised due to user's mistakes.
Image Uploader	IPFS	OneDrive	During campaign creation, the image inserted should be uploaded and url should be provided. IPFS saves image permanently using CID which removes the campaign-owner tracking and the risk of campaign image being removed by any cyber attack.

4.1.2 Identify External Entities

Over Specify Entities:

Initially, a broad list of potential entities is considered:

- Users (general individuals interacting with the platform)
- Donors (those contributing funds)
- Campaign Owners (those creating and managing campaigns)
- Blockchain Network (e.g., Ethereum)
- Wallet Providers (e.g., MetaMask)
- Regulatory Bodies
- Third-Party Services (e.g., payment gateways, analytics tools)
- Potential Investors

Refined Entities:

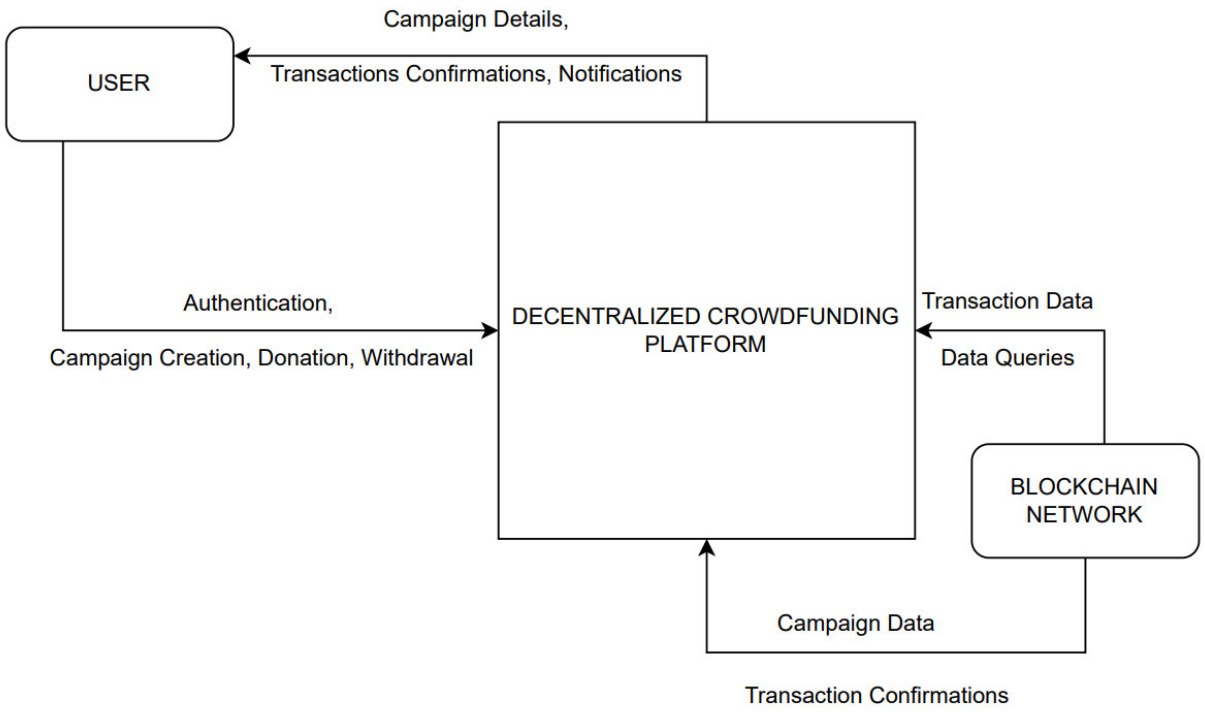
The list is refined to include only entities with direct and significant interactions with the system:

- **User:** Interacts with the platform to create campaigns, donate, or manage their profile. This includes both donors and campaign owners as sub-roles.
- **Blockchain Network (Ethereum):** Provides the decentralized infrastructure for recording transactions and executing smart contracts.
- **Wallet Provider:** Facilitates user authentication and transactions by connecting Ethereum wallets to the platform.
- **Administrator:** Manages the platform, including user support, system maintenance, and compliance oversight.

These refined actors clearly define the system's external environment and interactions.

Actor	Interaction with System
User	Creates campaigns, donates, manages profile
Blockchain Network	Processes transactions, executes contracts
Wallet Provider	Connects wallets, authorizes transactions
Administrator	Manages platform, supports users

4.1.3 Context Level Data Flow Diagram



4.1.4 Capture "shall" Statements

These are following.

- The system shall be technically feasible using blockchain technology (Ethereum) and smart contracts.
- The system shall be operationally feasible, integrating seamlessly into user workflows.
- The system shall be economically feasible, balancing development costs with benefits.
- The project shall be completed within a reasonable timeframe, guided by project management tools.

Specific Functionality:

- The system shall allow users to create crowdfunding campaigns with specified goals and timelines.
- The system shall enable users to connect their Ethereum wallets for transactions.
- The system shall allow donors to contribute to campaigns using Ethereum.
- The system shall track and display real-time campaign progress to users.
- The system shall allow campaign owners to withdraw funds once the campaign goal is met.
- The system shall ensure all transactions are recorded on the Ethereum blockchain for transparency.
- The system shall provide user authentication via wallet connections.

- The system shall comply with relevant legal and ethical standards for crowdfunding and cryptocurrency.

These statements define the core functionality required for the platform's success.

4.1.5 Allocate Requirements

Sr#	Initial Requirements	Use Case Name	Use Case ID
1	The system shall allow users to create crowdfunding campaigns with specified goals and timelines.	Create Campaign	UC-4
2	The system shall enable users to connect their Ethereum wallets for transactions.	Authenticate	UC-1
3	The system shall allow donors to contribute to campaigns using Ethereum.	Donate to Campaign	UC-3
4	The system shall track and display real-time campaign progress to users.	View Campaign Details	UC-2
5	The system shall allow campaign owners to withdraw funds once the campaign goal is met.	Withdraw Funds	UC-6
6	The system shall ensure all transactions are recorded on the Ethereum blockchain for transparency.	View Campaign Details	UC-2
7	The system shall provide user authentication via wallet connections.	Authenticate	UC-1
8	The system shall comply with relevant legal and ethical standards for crowdfunding and cryptocurrency.	Manage Profile	UC-5

4.1.6 Prioritize Requirements

Sr#	Priority	Initial Requirements	Use Case Name	Use Case ID
1	Highest	The system shall allow users to create crowdfunding campaigns with specified goals and timelines.	Create Campaign	UC-4
2	Highest	The system shall enable users to connect their Ethereum wallets for transactions.	Authenticate	UC-1
3	Highest	The system shall allow donors to contribute to campaigns using Ethereum.	Donate to Campaign	UC-3
4	Highest	The system shall track and display real-time campaign progress to users.	View Campaign Details	UC-2
5	High	The system shall allow campaign owners to withdraw funds once the campaign goal is met.	Withdraw Funds	UC-6
6	High	The system shall ensure all transactions are recorded on the Ethereum blockchain for transparency.	View Campaign Details	UC-2
7	Medium	The system shall provide user authentication via wallet connections.	Authenticate	UC-1
8	Medium	The system shall comply with relevant legal and ethical standards for crowdfunding and cryptocurrency.	Manage Profile	UC-5

Explanation:

- **Highest Priority:** Core features like campaign creation, wallet connection, donations, and progress tracking are critical for the platform's operation.
- **High Priority:** Features like fund withdrawal and transparency are essential but less urgent than the core functionality.
- **Medium Priority:** Additional authentication and legal compliance enhance the platform but can be refined after the initial launch.

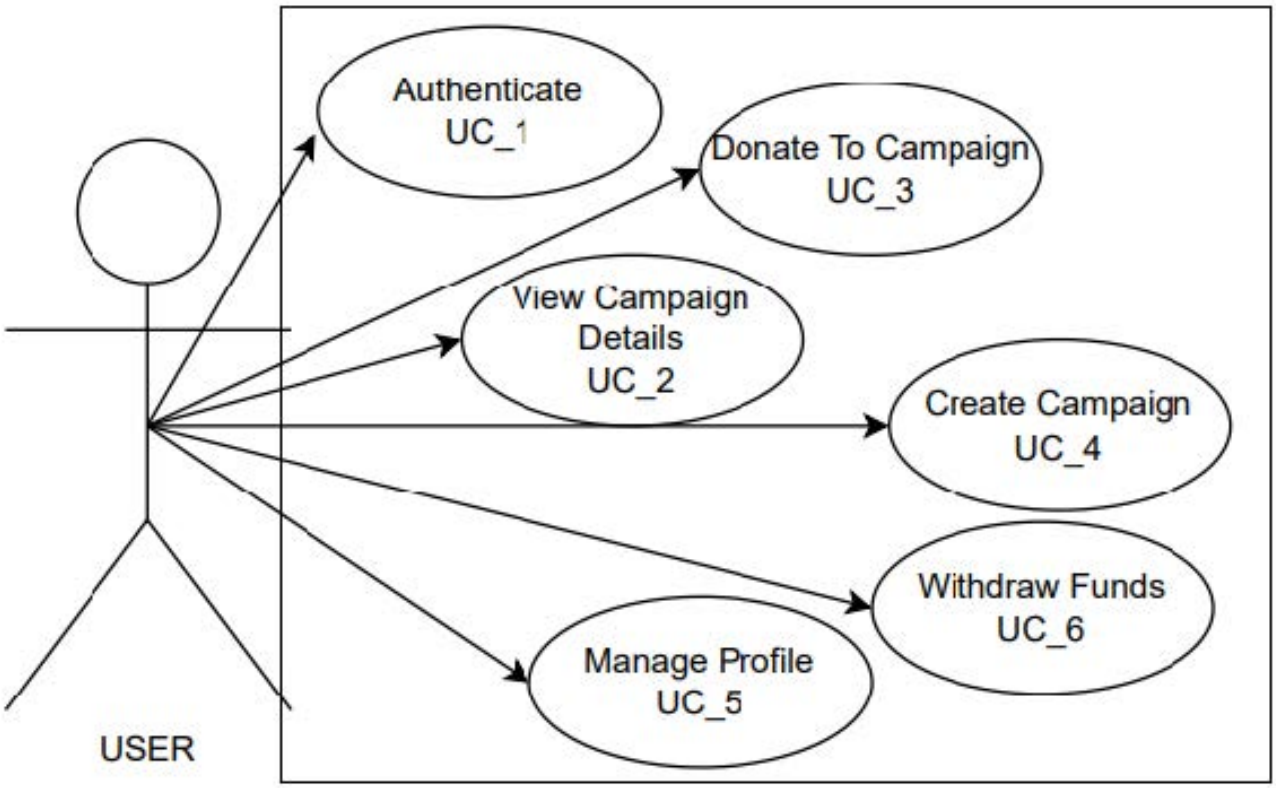
4.1.7 Requirements Trace-ability Matrix

Requirement ID	Requirement Description	Use Case ID	Design Element
REQ-1	Allow users to create crowdfunding campaigns with specified goals and timelines.	UC-4	CreateCampaign.jsx, Smart Contract
REQ-2	Enable users to connect their Ethereum wallets for transactions.	UC-1	ConnectWallet.jsx, Thirdweb SDK
REQ-3	Allow donors to contribute to campaigns using Ethereum.	UC-3	CampaignDetails.jsx, Smart Contract
REQ-4	Track and display real-time campaign progress to users.	UC-2	DisplayCampaigns.jsx, FundCard.jsx
REQ-5	Allow campaign owners to withdraw funds once the campaign goal is met.	UC-6	Withdraw.jsx, Smart Contract
REQ-6	Ensure all transactions are recorded on the Ethereum blockchain for transparency.	UC-2	Blockchain Network, Smart Contract
REQ-7	Provide user authentication via wallet connections.	UC-1	ConnectWallet.jsx, Thirdweb SDK
REQ-8	Comply with relevant legal and ethical standards for crowdfunding and cryptocurrency.	UC-5	Profile.jsx, Legal Compliance Module

Explanation:

- **Requirement ID:** Unique identifier for each requirement.
- **Requirement Description:** Concise statement of the requirement.
- **Use Case ID:** Links to the use case addressing the requirement.
- **Design Element:** Specific components (e.g., React JSX files or smart contracts) implementing the requirement.

4.2 High Level Use-case Diagram



CHAPTER # 5

5.1 Use-case Description

5.1.1 UC-1: Authenticate

Primary Actor: User

Brief Description: The User connects their cryptocurrency wallet to authenticate and access secure features on the dominate.pw platform, such as donating or creating campaigns.

Preconditions:

- The User has a compatible cryptocurrency wallet (e.g., MetaMask) installed and configured.

Post-Conditions:

- The User is authenticated, and their wallet is linked to the platform.

Main Success Scenario (Complete Normal Flow):

- The User clicks the "Connect Wallet" button on the dominate.pw interface.
- The system prompts the User to select a wallet provider.
- The User chooses their wallet and authorizes the connection request.
- The system verifies the wallet signature and establishes a secure connection.
- The system displays the User's wallet address, indicating successful authentication.
- The use case ends successfully.

Alternative Flows:

A1: The User cancels the wallet connection by closing the prompt or selecting "Cancel."

Exception:

E1: If the wallet connection fails due to network issues or misconfiguration, the system displays "Connection failed. Please check your wallet and try again."

5.1.2 UC-2: View Campaign Details

Primary Actor: User

Brief Description: The User views detailed information about a specific crowdfunding campaign on the dominate.pw platform, such as its title, description, funding goal, and progress.

Preconditions:

- The User has access to the platform (no authentication required for public campaigns).
- The campaign exists and is publicly visible.

Post-Conditions:

- The User has viewed the campaign details and can take further actions like donating or sharing.

Main Success Scenario (Complete Normal Flow):

- The User navigates to the campaign list on the dominate.pw homepage.
- The User selects a campaign to view its details.
- The system retrieves and displays the campaign's details, including title, description, funding goal, current progress, and deadline.
- The User reviews the campaign information and can choose to donate, share, or

return to the campaign list.
- The use case ends successfully.

Alternative Flows:
A1: The User applies filters (e.g., category, status) to the campaign list before selecting a campaign.
A2: The User accesses the campaign details directly via a shared link or URL.
Exception:
E1: If the campaign data fails to load due to network issues, the system displays "Failed to load campaign details. Please try again later."

5.1.3 UC-3: Donate To Campaign

Primary Actor: User
Brief Description: The User donates cryptocurrency funds to a selected campaign using their connected wallet.
Preconditions:
- The User is authenticated via their wallet.
- The campaign is active and accepting donations.
- The User has sufficient funds in their wallet.

Post-Conditions:
- The donation is processed, and the campaign's funding progress is updated.

Main Success Scenario (Complete Normal Flow):
- The User selects "Donate" on the campaign details page.
- The system displays a donation form where the User can specify the amount in ETH.
- The User enters the desired amount and clicks "Confirm Donation."
- The system prompts the User to confirm the transaction via their wallet.
- The User approves the transaction in their wallet.
- The system processes the donation via the smart contract and updates the campaign's funding status.
- The system displays "Donation Successful" and shows the updated funding progress.
- The use case ends successfully.

Alternative Flows:
A1: The User cancels the donation before confirming the transaction.
Exception:
E1: If the transaction fails (e.g., insufficient funds or network issues), the system displays "Donation failed. Please try again."

5.1.4 UC-4: Create Campaign

Primary Actor: User
Brief Description: The User creates a new crowdfunding campaign by providing details like title, description, funding goal, and deadline.

Preconditions:
- The User is authenticated via their wallet.
- The User agrees to the platform's terms and conditions.

Post-Conditions:
- A new campaign is created and published on the platform.

Main Success Scenario (Complete Normal Flow):
- The User clicks "Create Campaign" on the dominate.pw interface.
- The system displays a form requiring campaign details (e.g., title, description, funding goal, deadline).
- The User fills in the required information and clicks "Submit."
- The system validates the input data (e.g., ensures funding goal is positive).
- The system creates the campaign via the smart contract and publishes it on the platform.
- The system displays "Campaign Created Successfully" and redirects the User to the campaign's details page.
- The use case ends successfully.

Alternative Flows:
A1: The User sees a pop up error if some inserted data is invalid.
Exception:
E1: If the input data is invalid (e.g., missing fields), the system displays "Please correct the errors and try again."

5.1.5 UC-5: Manage Profile
Primary Actor: User
Brief Description: The User profile information is just their wallet address on the dominate.pw platform.
Preconditions:
- The User is authenticated via their wallet.

Post-Conditions:
- The User's created campaigns information is available with transaction records.

Main Success Scenario (Complete Normal Flow):
- The User navigates to their profile page on the dominate.pw platform.
- The system displays the current campaigns created by user.
- The use case ends successfully.

Alternative Flows:
A1: The User cancels the changes by clicking "Cancel" before saving.
Exception:
E1: If the data fetch fails due to offline node, the system displays "Update failed. Please try again."

5.1.6 UC-6: Withdraw Funds
Primary Actor: User
Brief Description: The User withdraws funds from a campaign they own, transferring the raised cryptocurrency to their connected wallet.

Preconditions:

- The User is authenticated via their wallet.
- The User is the owner of the campaign.
- The campaign has sufficient funds available for withdrawal.

Post-Conditions:

- The specified amount is transferred to the User's wallet, and the campaign's balance is updated.

Main Success Scenario (Complete Normal Flow):

- The User selects "Withdraw Funds" on their campaign's details page.
- The system displays a withdrawal form where the User can specify the amount to withdraw.
- The User enters the amount and clicks "Confirm Withdrawal."
- The system prompts the User to confirm the transaction via their wallet.
- The User approves the transaction in their wallet.
- The system processes the withdrawal via the smart contract and updates the campaign's balance.
- The system displays "Withdrawal Successful" and shows the updated balance.
- The use case ends successfully.

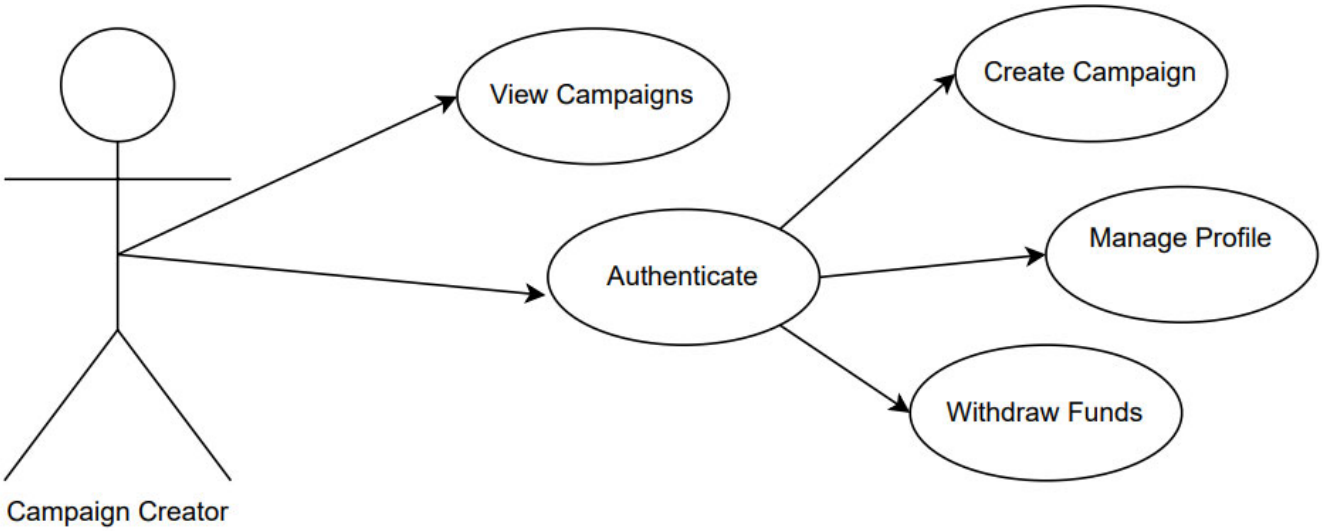
Alternative Flows:

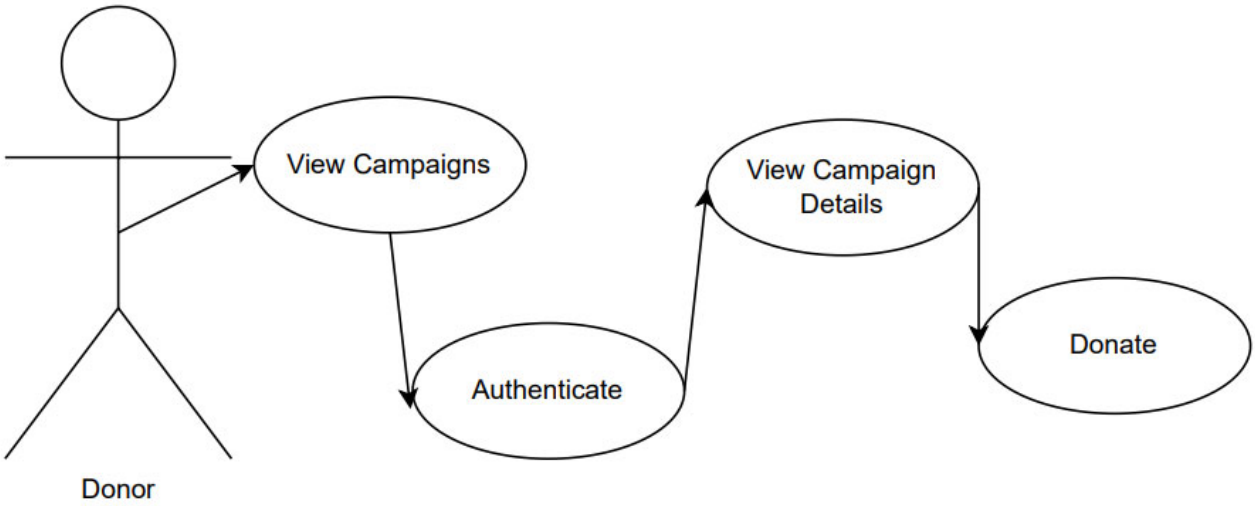
A1: The User cancels the withdrawal before confirming the transaction.

Exception:

E1: If the withdrawal fails (e.g., insufficient funds or network issues), the system displays "Withdrawal failed. Please try again."

5.1.6 Use-case (Refined)

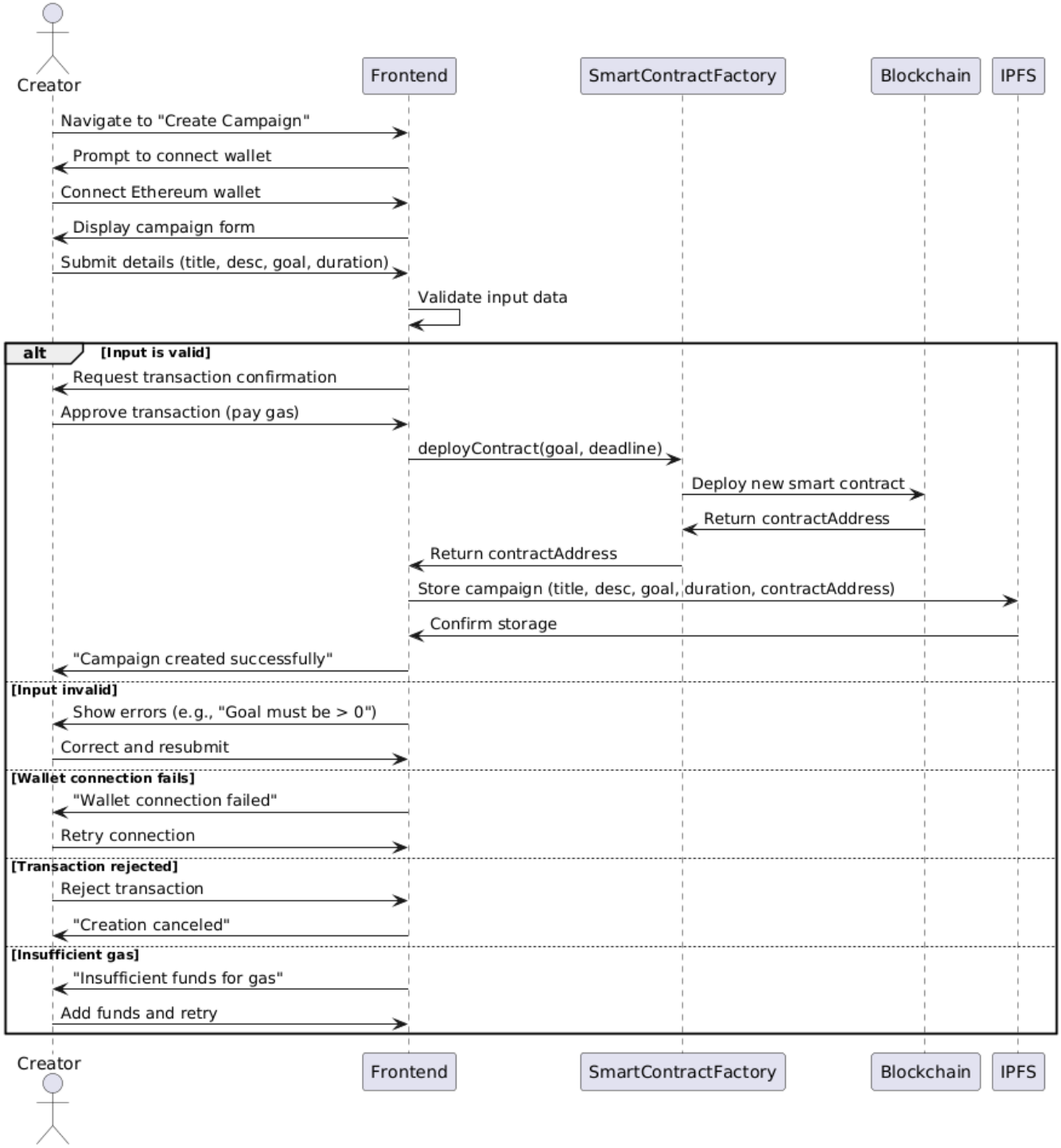




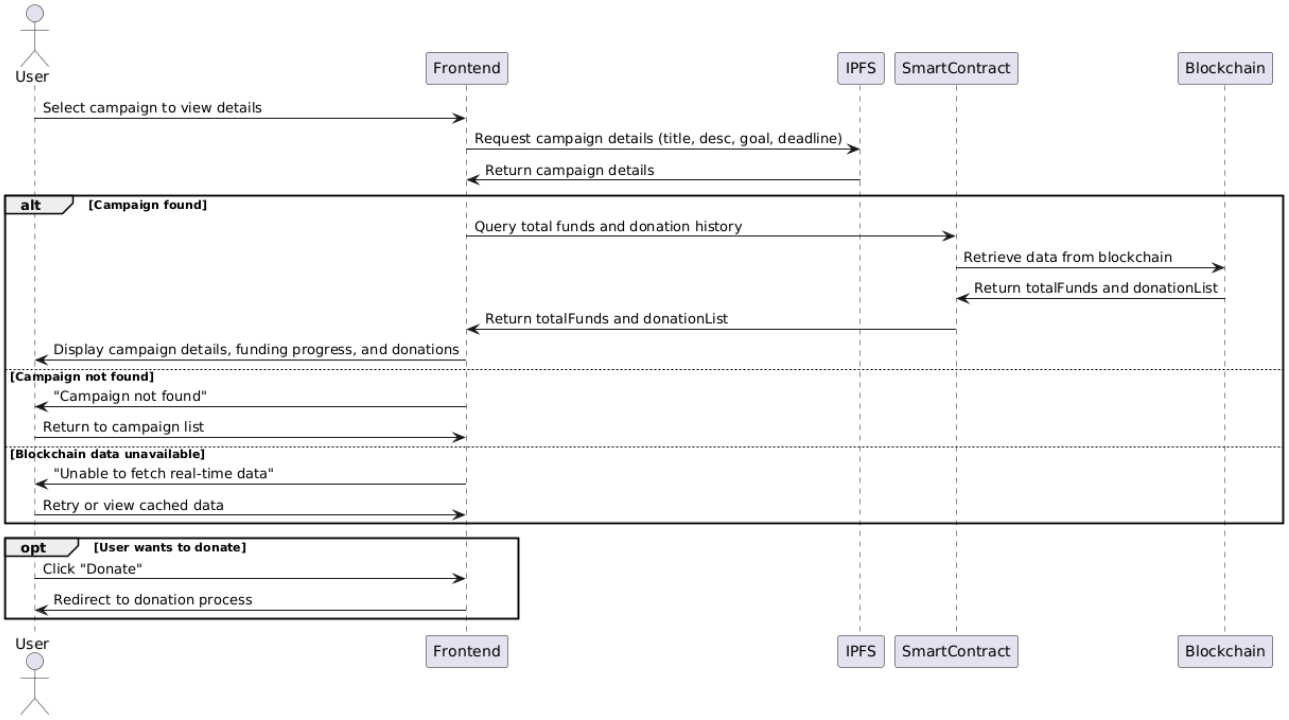
CHAPTER # 6

6.1 Sequence Diagram

6.1.1 Create Campaign

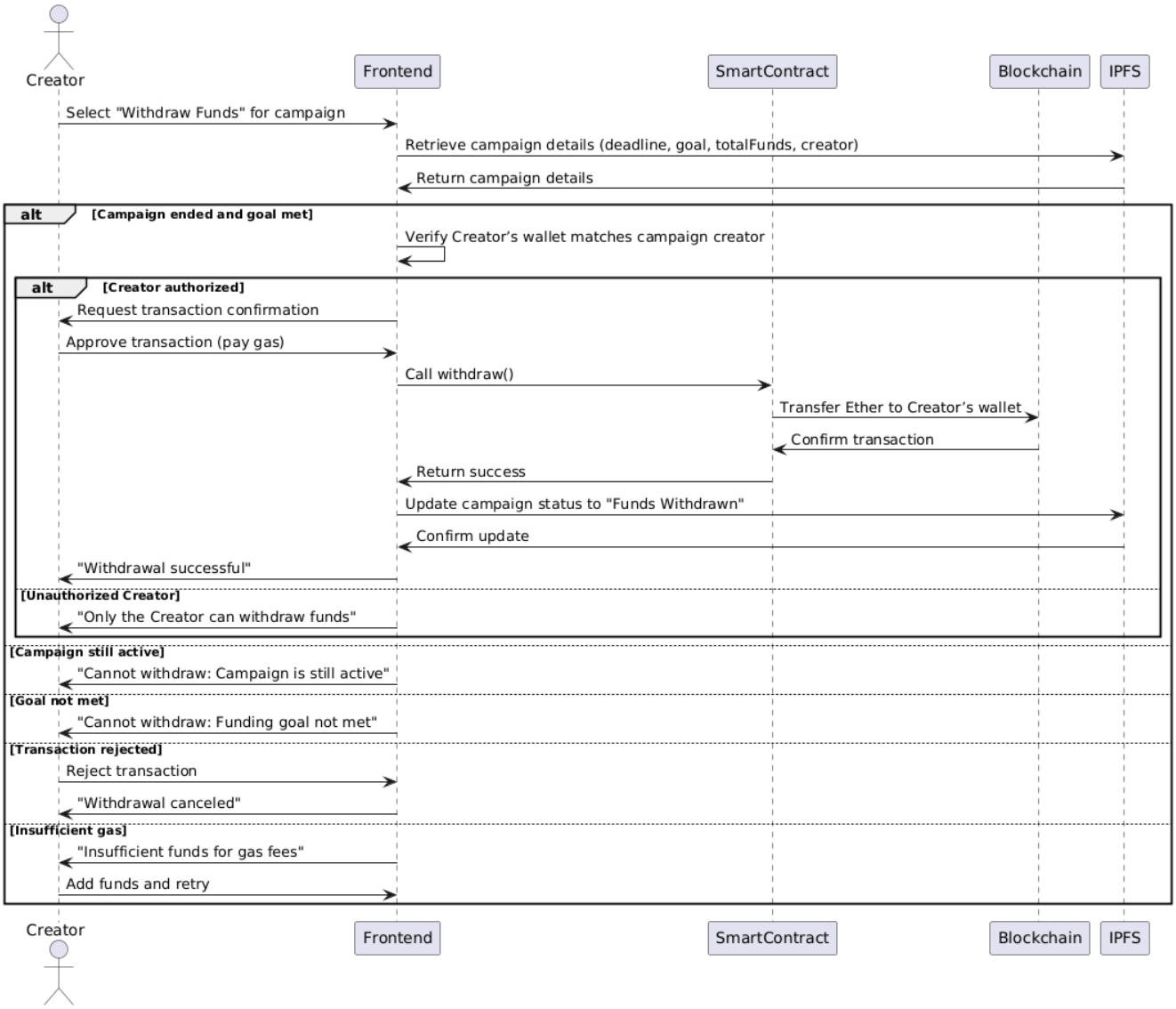


6.1.2 View Campaign Details



6.1.3 Donate to Campaign

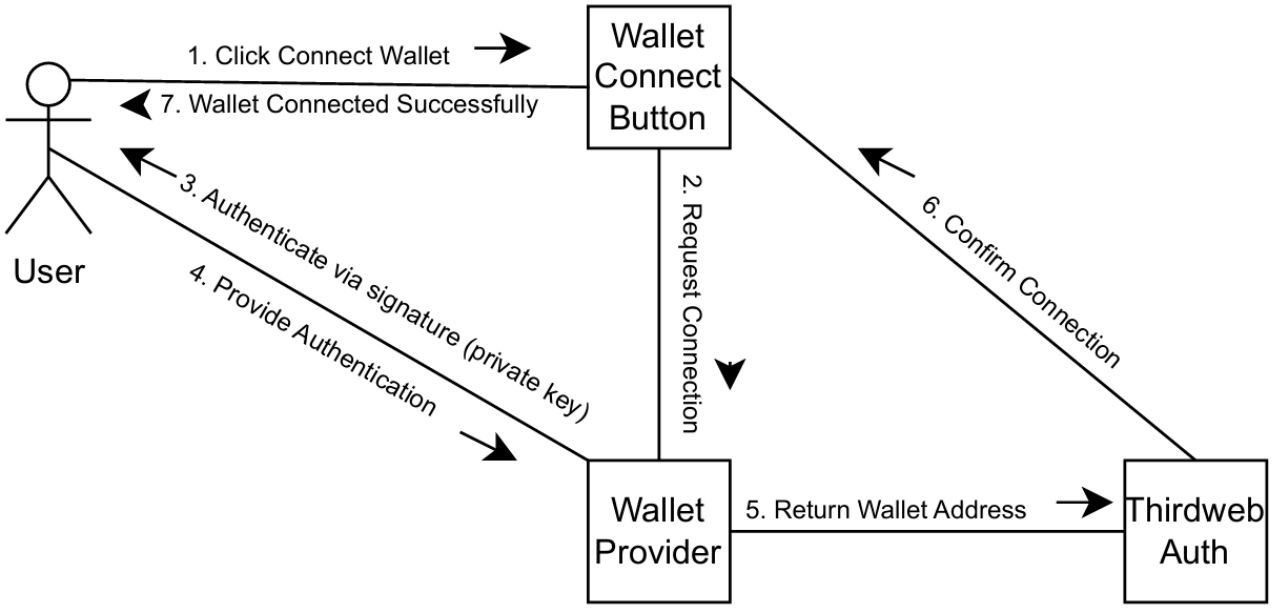
6.1.4 Withdraw Funds



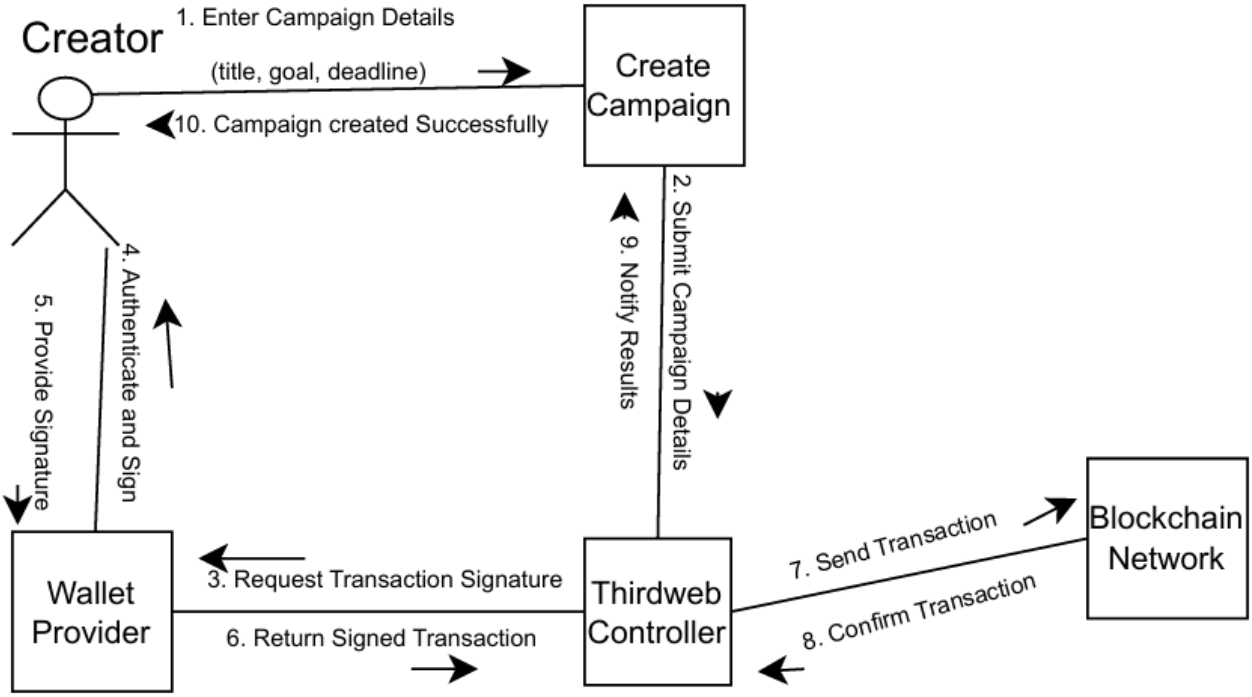
CHAPTER # 7

7.1 Collaboration Diagram

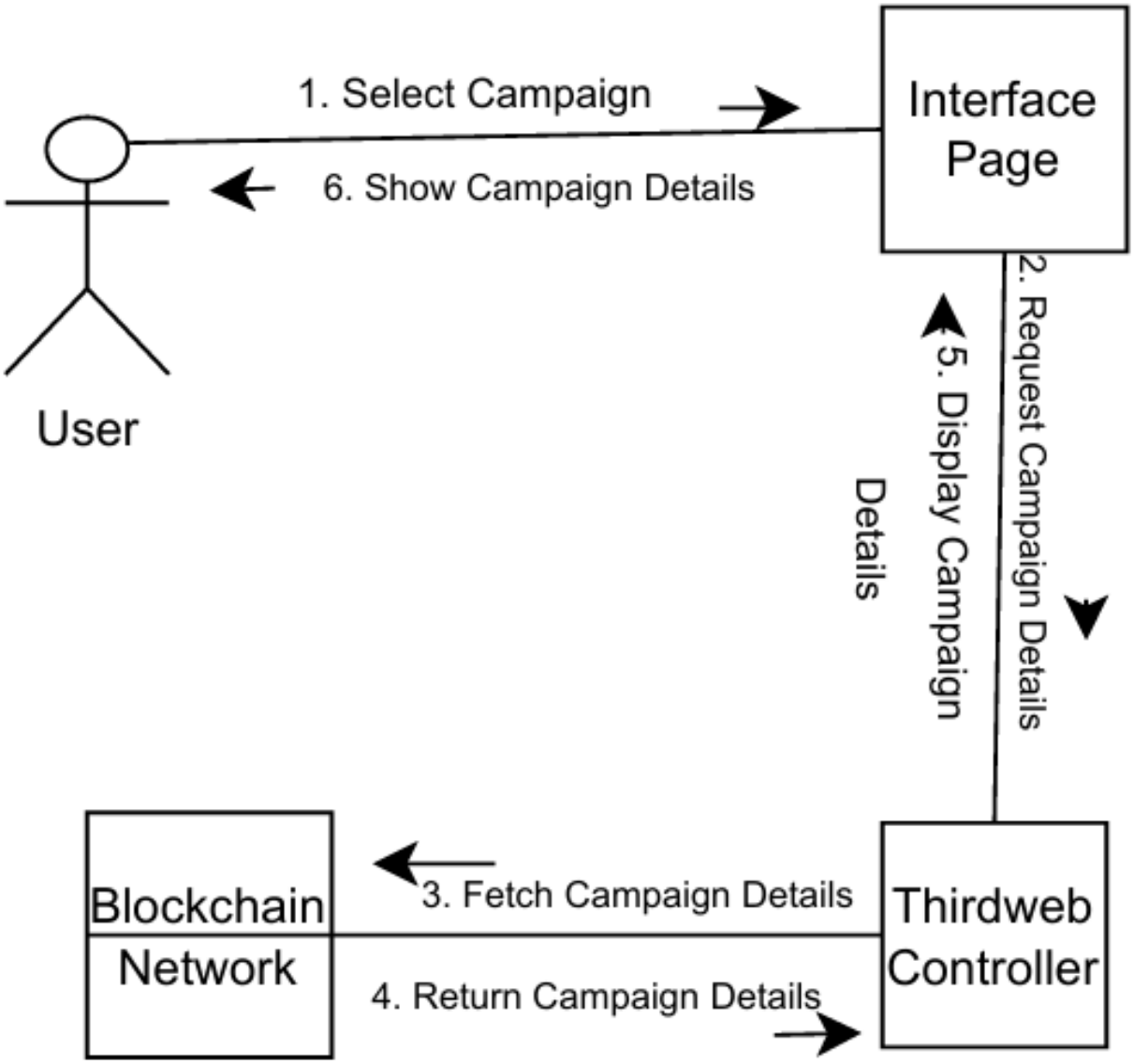
7.1.1 Connect Wallet



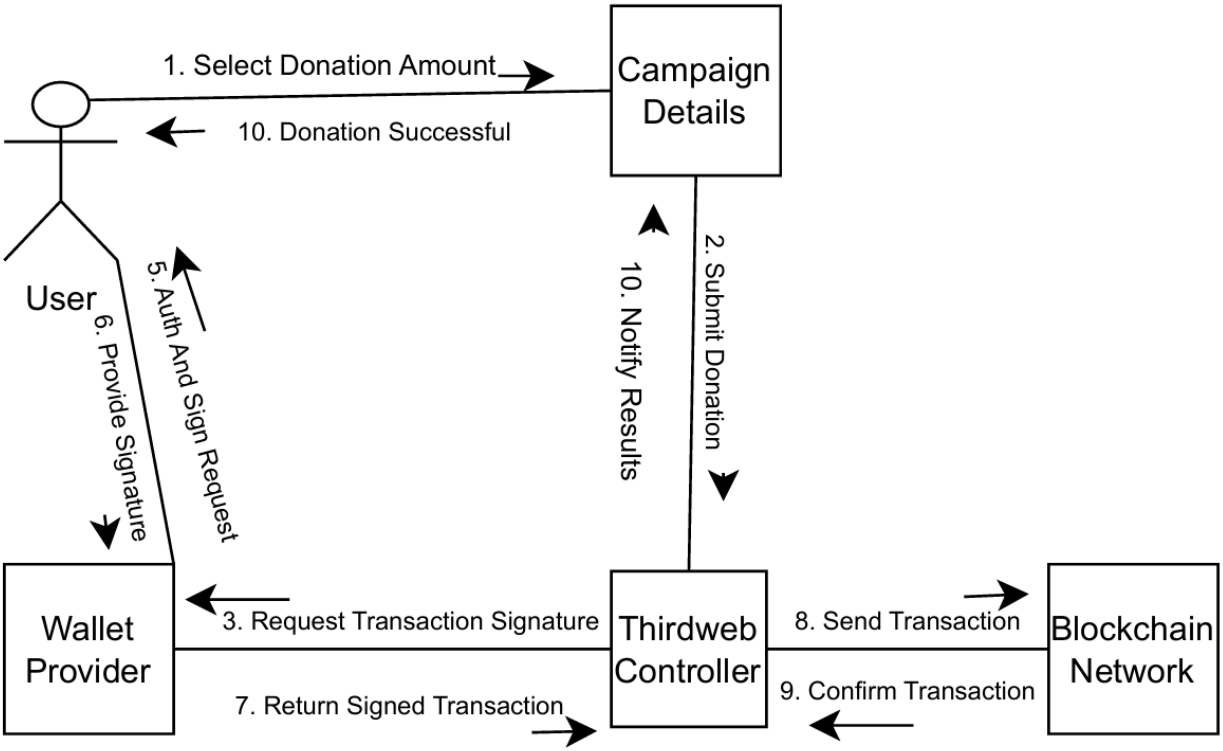
7.1.2 Create Campaign



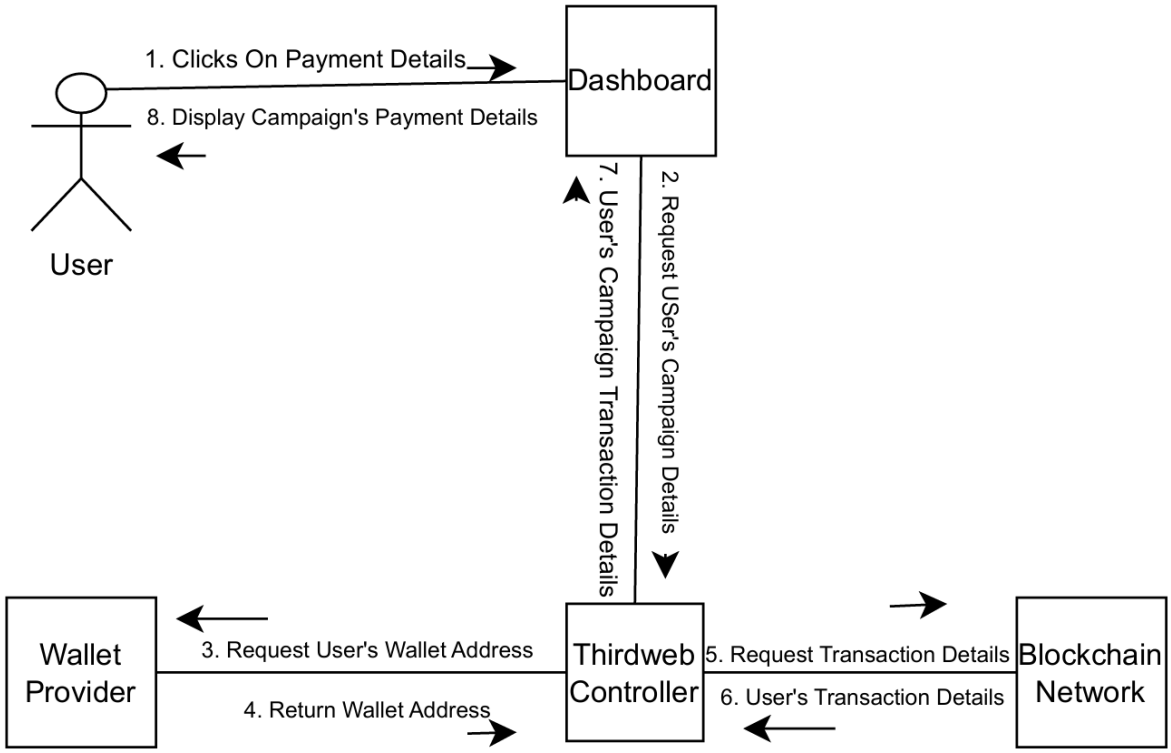
7.1.3 View Campaign Details



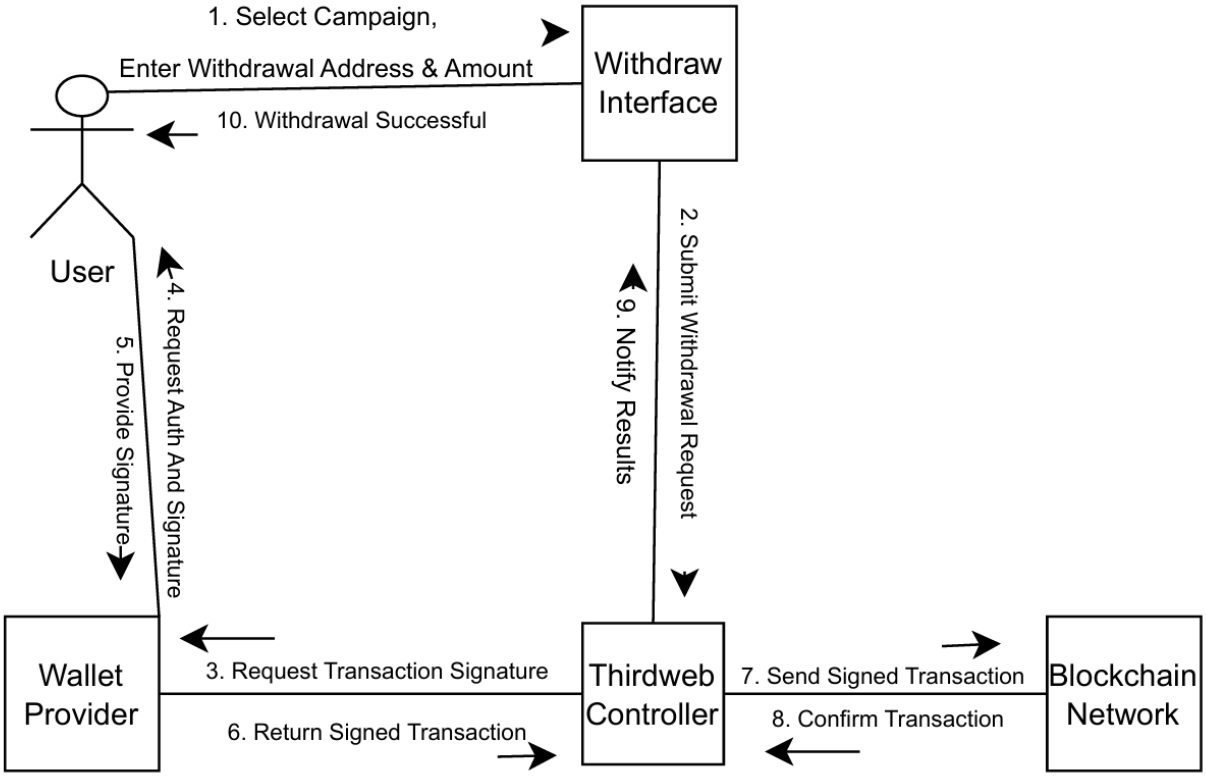
7.1.4 Donate To Campaign



7.1.5 Payment Details

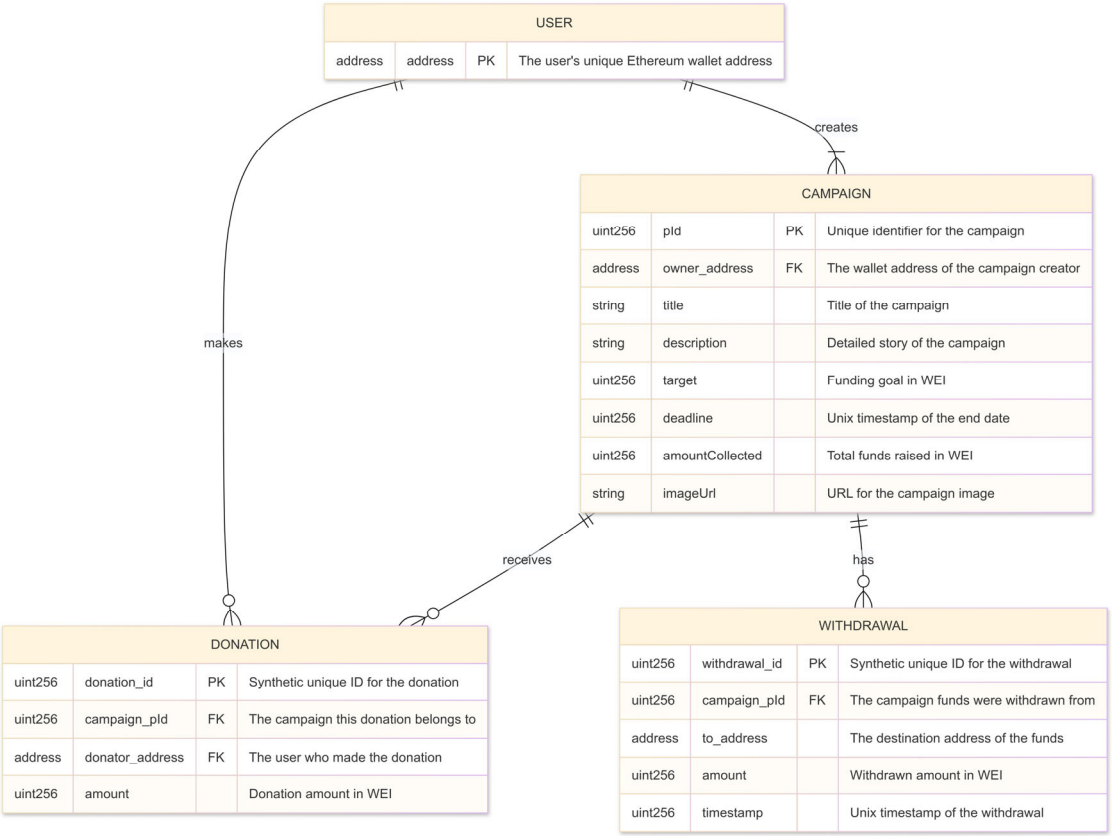


7.1.6 Withdraw Funds



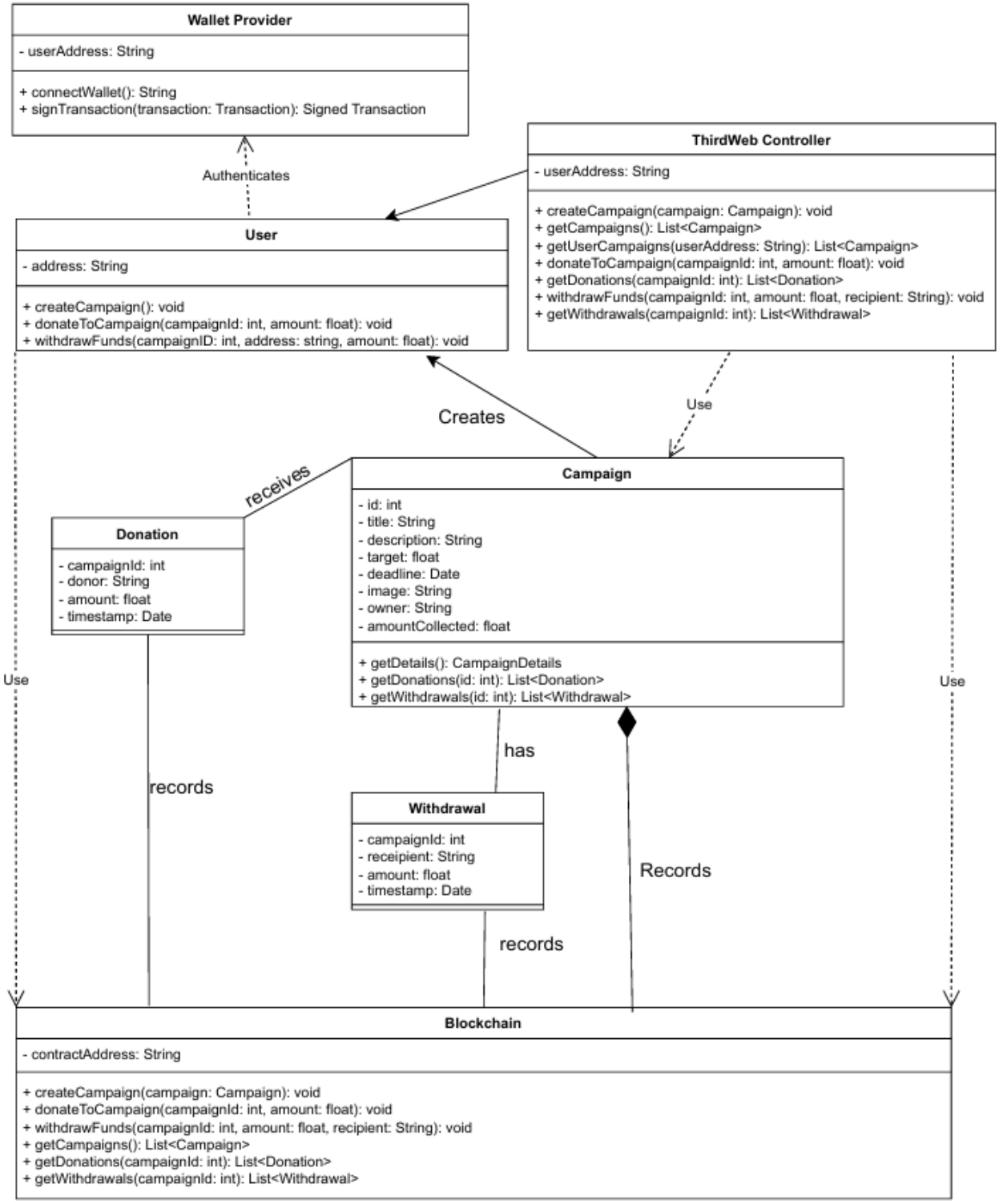
CHAPTER # 8

8.1 Data Diagram



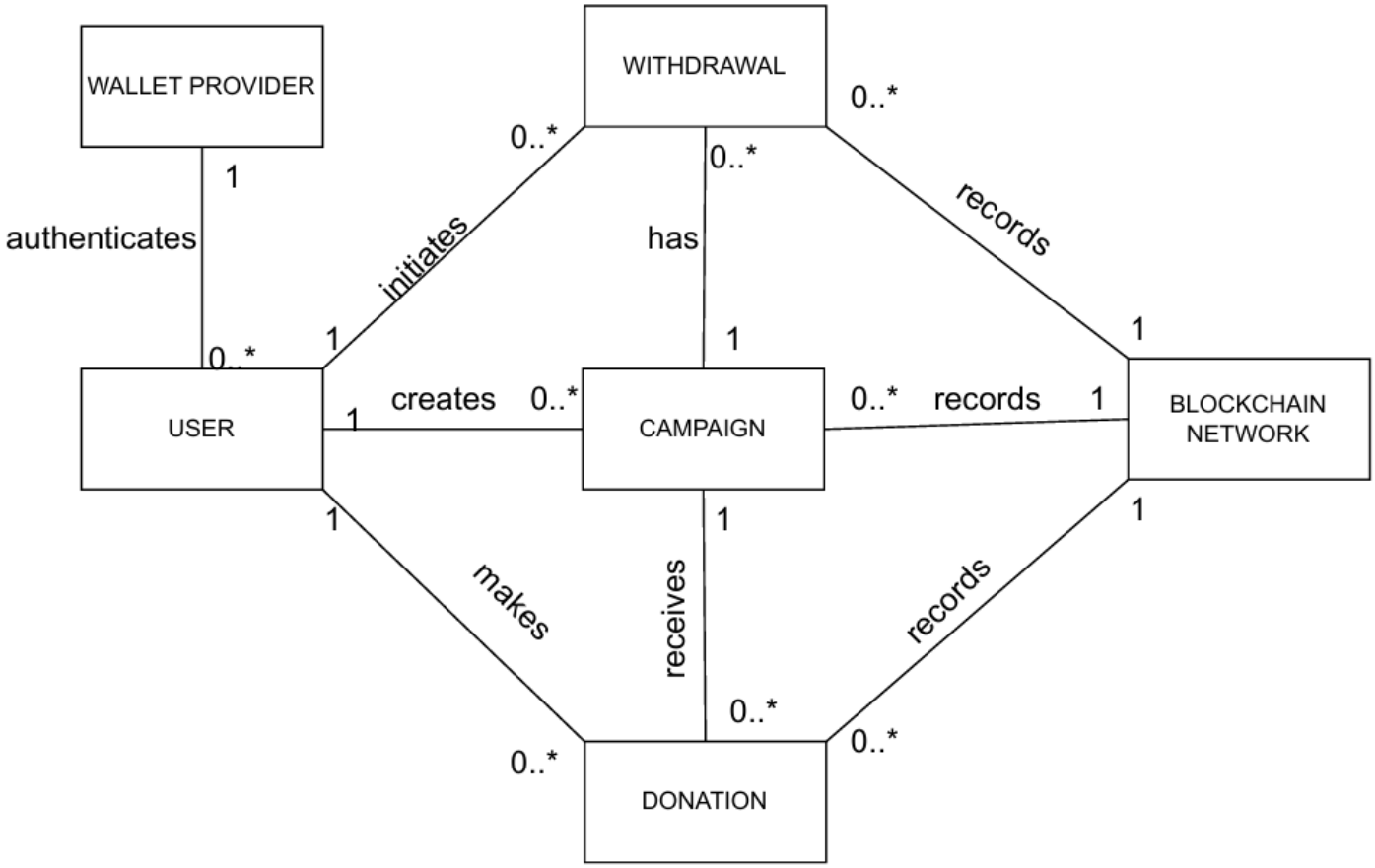
CHAPTER # 9

9.1 Class Diagram



CHAPTER # 10

10.1 Domain Model



Entities

1. **User**

Individuals who interact with the platform, such as campaign creators and donors.

2. **Campaign**

Crowdfunding projects created by users, containing details like title, funding goal, and deadline.

3. **Donation**

Contributions made by users to campaigns, including amount and donor information.

4. **Withdrawal**

Requests by campaign owners to withdraw funds once goals are met.

5. **Blockchain Network**

The Ethereum blockchain (Holesky testnet) where all transactions are recorded.

6. **Wallet Provider**

Services like MetaMask that authenticate users and facilitate transactions.